Introduction to Football Data & Analytics

CSE490 (D) | Lecture 1

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BANGLADESH FOOTBALL FEDERATION

Lecture Outline

Part I: Course Overview

1. Objectives
2. Course content
3. Assessment structure
4. Research project
5. Tools

Part II: Football Data

6. StatsBomb data structure
7. Competitions, matches, and events
8. Visualizing events
9. StatsBomb 360 data
10. Event data vs. tracking data

Course Objectives

By the end of this course, students will be able to:

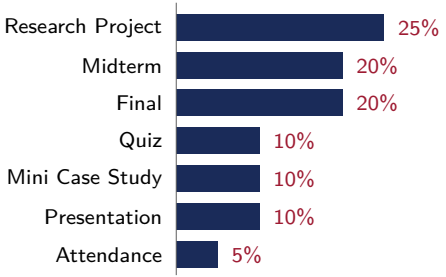
- Introduce **research-oriented football analytics** using computing and statistical methods
- Develop **analytical, modeling, and visualization** skills using real-world data
- Foster **independent research**, critical thinking, and lifelong learning
- Strengthen **ethical reasoning**, teamwork, and professional communication

Course Content

Module	Lessons
Data Foundations & Visualization	Introduction to Event Data & KPIs; Visualizing Football; Passing Networks & Team Structure
Statistical Models & Expected Goals	Choosing & Fitting Models; Advanced Models; Building an xG Model
Player & Action Valuation	Scouting and Player Profiles; Valuing Specific Actions
Uncertainty, Prediction & Trends	Randomness in Football; Team-Level KPIs & Trend Analysis
Space Control & Player Movement	Pitch Control Tools; Off-Ball Movement & Positioning
Physical Data & Clustering Techniques	Physical Performance Metrics; Clustering in Football Analytics

Assessment Structure

Assessment	Weight
Attendance	5%
Quiz	10%
Midterm	20%
Research Project	25%
Mini Case Study	10%
Final	20%
Research Presentation	10%
Total	**100%**



Research Project

Nature

- Group project (**4 students**)
- Dataset: StatsBomb / Wyscout / other open football data

Deliverables

1. Research proposal
2. Literature review
3. Methodology & modeling
4. Results & discussion
5. Ethical considerations
6. Final report & presentation

Research Project: Grading Rubric

Component	Weight
Problem Formulation	15%
Literature Review	15%
Methodology	20%
Analysis & Validation	25%
Ethics & Limitations	10%
Communication	15%
Total	**100%**

Where the marks are

Methodology and **Analysis & Validation** together carry **45%** of the project grade. Rigorous modeling and honest validation matter most.

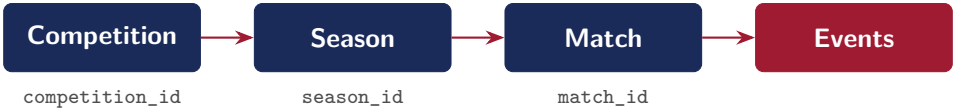
Tools

- **Language:** Python
- **Open datasets:** StatsBomb / Wyscout
- **Libraries:**
  - `pandas`, `numpy`: data handling and numerics
  - `matplotlib`, `mplsoccer`: visualization and pitch plotting
  - `statsbombpy`: access to StatsBomb open data

Note

We will mostly use **Google Colab**, but it is better if you also have **conda** or **VS Code** installed locally.

StatsBomb: Data Structure



How to access data

- StatsBomb assigns **unique IDs** at each level
- Each competition, season, and match has its own ID

Example

Competition:	FIFA World Cup (43)
Season:	2018 (106)
Match:	England vs Colombia (7580)

Follow the course notebook: Google Colab notebook

StatsBomb: Competitions

Loading competitions as a **DataFrame**; this provides high-level information such as:

- Competition name / country
- Season
- Number of matches
- Gender
- Availability of **360 data**

#	Column	Non-Null Count	Dtype
0	competition_id	75 non-null	int64
1	season_id	75 non-null	int64
2	country_name	75 non-null	object
3	competition_name	75 non-null	object
4	competition_gender	75 non-null	object
5	competition_youth	75 non-null	bool
6	competition_international	75 non-null	bool
7	season_name	75 non-null	object
8	match_updated	75 non-null	object
9	match_updated_360	57 non-null	object
10	match_available_360	11 non-null	object
11	match_available	75 non-null	object

StatsBomb: Matches

Loading matches for a competition as a **DataFrame**; this provides match-level information such as:

- Home team & away team
- Match score
- Stadium
- Match stage
- Referee
- Team managers
- Match date

#	Column	Non-Null Count	Dtype
0	match_id	64 non-null	int64
1	match_date	64 non-null	object
2	kick_off	64 non-null	object
3	competition	64 non-null	object
4	season	64 non-null	object
5	home_team	64 non-null	object
6	away_team	64 non-null	object
7	home_score	64 non-null	int64
8	away_score	64 non-null	int64
9	match_status	64 non-null	object
10	match_status_360	64 non-null	object
11	last_updated	64 non-null	object
12	last_updated_360	64 non-null	object
13	match_week	64 non-null	int64
14	competition_stage	64 non-null	object
15	stadium	64 non-null	object
16	referee	64 non-null	object
17	home_managers	64 non-null	object
18	away_managers	64 non-null	object
19	data_version	64 non-null	object
20	shot_fidelity_version	0 non-null	object

StatsBomb: Events

Loading events for a match as a **DataFrame**. Each row represents **one on-ball action**.

Examples of events:

- Pass, Shot, Foul, Duel, Clearance

Information stored in each event:

- Event type
- Time (minute, second)
- Event location on the pitch
- Player and team involved
- Event-specific details (e.g., pass type, shot outcome)

StatsBomb: Events and Missing Values

Important observation

Most columns of the events DataFrame contain `NaN` values.

This is expected because:

- Each event type only uses a **subset of columns**
- Columns not relevant to an event remain empty

StatsBomb: Events and Missing Values

```
Data columns (total 83 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   50_50                           4 non-null      object
 1   bad_behaviour_card              3 non-null      object
 2   ball_receipt_outcome            160 non-null    object
 3   ball_recovery_recovery_failure  13 non-null     object
 4   block_deflection                2 non-null      object
 5   block_offensive                 1 non-null      object
 6   carry_end_location              851 non-null    object
 7   clearance_aerial_won            4 non-null      object
 8   counterpress                    108 non-null    object
 9   dribble_nutmeg                  2 non-null      object
```

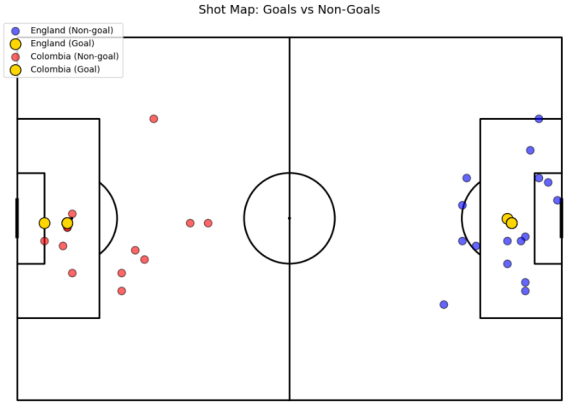
StatsBomb: Extracting Shot Events

We can extract very specific (non-`NaN`) information for a **"Shot"** event.

Event Type	Player	Team	Minute	Location (x, y)	Body Part	Shot Type	Outcome	xG
Shot	Ashley Young	England	5	(115.0, 18.0)	Right Foot	Free Kick	Saved	0.0098
Shot	Raheem Sterling	England	7	(112.0, 54.0)	Right Foot	Open Play	Blocked	0.0382
Shot	Raheem Sterling	England	12	(98.0, 37.0)	Right Foot	Open Play	Blocked	0.0451
Shot	Harry Kane	England	15	(119.0, 36.0)	Head	Open Play	Off Target	0.6251
Shot	Juan Cuadrado	Colombia	21	(97.0, 56.0)	Right Foot	Open Play	Blocked	0.0218

StatsBomb: Visualizing Shots

We can draw a standard-size football pitch and visualize events using **mplsoccer's** `Pitch`.



Illustrative shot map on a 120×80 StatsBomb pitch

StatsBomb: 360 Data

StatsBomb offers **360 data**, which tracks not only the location of an event but also the **locations of players** in the camera's view.

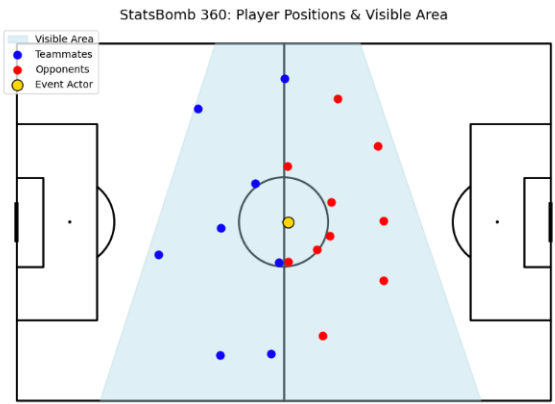
#	Column	Non-Null Count	Dtype
0	id	41562 non-null	object
1	visible_area	41562 non-null	object
2	match_id	41562 non-null	int64
3	teammate	41562 non-null	bool
4	actor	41562 non-null	bool
5	keeper	41562 non-null	bool
6	location	41562 non-null	object

Player locations at the moment of the event:

	x	y	teammate	actor	keeper
0	31.885449	47.289130	True	False	False
1	40.762566	14.657225	True	False	False
2	45.765232	69.890950	True	False	False
3	45.923999	41.324219	True	False	False
4	53.649592	31.368630	True	False	False
5	57.221077	69.484720	True	False	False
6	58.963536	49.056717	True	False	False
7	60.159949	7.849366	True	False	False
8	60.952056	27.553933	False	False	False

StatsBomb: 360 Data in Practice

Question: What can you observe from this pitch image?



Match details

Serbia vs Switzerland
Match date: 2022-12-02
Serbia formation: **3-4-1-2**
Switzerland formation: **4-2-3-1**

These are also extracted using `match_id` and keys such as `home_team`, `away_team`, `match_date`, `type`, and `tactics`.

Event Data vs. Tracking Data

Event data

- Records **on-ball actions** (passes, shots, tackles, fouls)
- Logged **only when an event occurs**
- **Sparse** data; many `NaN` columns (not all fields apply to all events)
- Used for **xG**, shot maps, pass maps, KPIs

Tracking data

- Records **continuous positions** of all players and the ball
- Collected **every frame** (e.g., 10 times per second)
- **Dense**, large datasets
- Used for **pressing**, team shape, space control, movement analysis

Event vs. Tracking: At a Glance

	Event data	Tracking data
Records	On-ball actions	Positions of all players and the ball
Frequency	When an event occurs	Every frame (e.g., 10 Hz)
Density	Sparse (many `NaN`)	Dense, very large
Typical uses	xG, shot maps, pass maps, KPIs	Pressing, team shape, space control, movement

References

- **Blog:** Welcome to Soccermatics – Soccermatics documentation
- **Course:** Mathematical Modelling of Football (YouTube)
- **Book:** D. Sumpter, *Soccermatics*

Questions?

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